



OPEN Shifts in task absorption during decision-making episodes

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In today's rapidly evolving technological landscape, this study investigates the impact of new technology on organizational work dynamics. By integrating the Job Demands-Resources theory and the Episodic Process Model, we examine the cognitive mechanisms that influence task performance during decision-making episodes. Our research focuses on episodic job demands and resources, emphasizing the mediating role of task absorption in the relationship between these factors and task performance. Our findings reveal that episodic job resources positively affect task absorption and performance, while job demands moderate the relationship between job resources and task absorption. Employing the Beer Distribution Game alongside neuroscience-based eye-tracking techniques, we analyze visual attention dynamics during decision-making episodes, providing novel insights into the interplay between task absorption and task performance. Theoretically, our study highlights the significance of task absorption in understanding how episodic job resources impact decision-making performance. Practically, our results advocate for the implementation of decision-making-focused skills development through physiological measures such as neurofeedback training programs. This research underscores the importance of individual time management in enhancing task performance, contributing to a nuanced understanding of job resources, job demands, and task absorption in the context of technological transformation.

Keywords Task absorption, Decision-making, Episodes, Subjective, Behavioral, Physiological

The dynamics of job demands within organizational contexts are characterized by rapid fluctuations within episodes, resulting in diverse outcomes ranging from burnout to engagement^{1,2}. These fluctuations, particularly pronounced during decision-making episodes, significantly shape individuals' experiences at work³. Rapid shifts in demands can either overwhelm individuals, leading to burnout or deeply engage them, fostering a sense of fulfillment and accomplishment. Decision-making episodes, marked by their high stakes and complexity, exemplify this phenomenon as managers navigate through a myriad of options and considerations within limited time frames³. Thus, investigating the interplay between task absorption, job demands, and decision-making episodes offers invaluable insights into optimizing organizational performance while mitigating the risk of burnout⁴.

The landscape of organizational work is currently undergoing a profound metamorphosis driven by technological advancements. Work experiences are increasingly structured around episodes, delineated as cohesive time segments aimed at achieving work-related objectives⁵. With technology assuming responsibility for routine task automation, human operators are compelled to delve deeper into the cognitively demanding facets of their work, necessitating extensive information processing⁶. Task absorption, recognized as a cornerstone of task performance during work episodes, has emerged as a focal point among scholars^{7–9}. This state of deep immersion spans across various professions, from artists and athletes to scientists and managers, indicating heightened performance characterized by intense focus and subjective perception of time passing swiftly^{1,10}. The ability of managers to achieve optimal outcomes in decision-making tasks hinges on their level of task absorption¹¹. Such tasks, integral to managerial workflow, demand full immersion to effectively navigate complexities¹². Supply chain managers, for instance, must maintain unwavering focus to manage inventory effectively, as interruptions can compromise decision quality¹¹. However, not all tasks necessitate deep absorption. Some managerial duties, such as drafting employee evaluations, may be adequately performed without sustained absorption if overall effectiveness is maintained. Yet, the quality of certain tasks, particularly those involving critical decisions like supply chain management, cannot be easily compensated for by subsequent episodes. Thus, task characteristics play a pivotal role in determining the level of absorption required during work episodes.

Despite its significance, the relationship between task characteristics and task absorption remains relatively unexplored. To address this gap, our research delves into the sub-episodes of decision-making within tasks,

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examining how individuals process information and make decisions. Employing the neuroscientific technique of eye-tracking, we aim to measure task absorption by analyzing physiological indicators such as pupil diameter¹³. Drawing upon the Job Demands and Resources (JD-R) theory¹, we explore how episodic job resources influence task performance through the mediating mechanism of task absorption. Our study contributes to three key areas: elucidating the role of task absorption in mediating the impact of job resources on task performance, integrating objective physiological measurements into organizational research, and providing practical insights for managerial training and decision-making processes.

In the subsequent sections, we present a comprehensive review of the literature, formulate hypotheses, and elucidate our method, which employs experimental design to test our propositions. Finally, we discuss our findings, implications, and conclusions, thereby offering valuable insights into the dynamics of work episodes and decision-making processes within organizations.

Task absorption

Task absorption is most prominently associated with the concept of flow, as proposed by¹⁴. Flow occurs when the challenge level of a task matches an individual's skill level, resulting in a state of optimal experience and heightening focus. Other theories, such as self-determination theory¹⁵ also emphasize the importance of intrinsic motivation and autonomy in achieving task absorption¹⁰. define task absorption as a mental state characterized by solid attention. Additionally¹, states that task absorption is a positive state of mind when the individual is so immersed that time seems to fly.

Task absorption has practical applications in various domains. For example, in education, educators can design learning experiences that promote flow, fostering student engagement and learning¹⁶. Athletes often seek flow states to optimize their performance in sports and performance, and coaches can use flow principles to enhance training regimens¹⁶. In the workplace, companies can create work environments that support flow by providing challenging tasks, autonomy, and opportunities for skill development¹.

Following the existing literature, several factors contribute to the likelihood of experiencing task absorption: (1) Skill-Task Balance: As mentioned by¹⁴, an ideal balance between skills and task difficulty is crucial. When tasks are too easy or too tricky, task absorption is less likely to occur¹⁴; (2) Clear Goals: Well-defined goals and objectives provide a sense of purpose and direction, facilitating task absorption¹⁷; (3) Immediate Feedback: Timely and relevant feedback during the task can help individuals stay engaged and immersed¹⁴; and (4) Autonomy: The ability to make choices and have a degree of control over the task can enhance task absorption¹⁵.

Task absorption has been associated with numerous positive outcomes: (1) Enhanced Performance: Individuals in a state of task absorption often exhibit higher levels of performance and productivity¹; (2) Increased Creativity: Flow can stimulate creative thinking and problem-solving abilities¹⁴; (3) Improved Well-being: Experiencing task absorption is linked to increased feelings of happiness, satisfaction, and reduced stress⁸; (4) Long-term Engagement: Individuals who frequently experience task absorption are more likely to stay engaged to their pursuits over time⁸. Task absorption, commonly called flow, is a psychological state associated with deep immersion in a task, leading to enhanced performance, well-being, and long-term engagement. Understanding the antecedents and outcomes of task absorption can have significant implications in education, the workplace, sports, and other areas of human endeavor. Future research may explore strategies for intentionally inducing and sustaining task absorption and its impact on various aspects of human life.

JD-R theory

The JD-R theory is one of the most cited theories in applied psychology. The theory argues that all job characteristics can be classified into job demands and resources. Job demands refer to those physical, psychological, social, or organizational aspects of work that require sustained physical and mental effort and, therefore, are associated with specific physiological and psychological costs¹⁸. Examples are emotional and cognitive demands. Job resources refer to those physical, psychological, social, or organizational aspects of work that are functional in achieving job goals, reducing job demands and associated physiological and psychological costs, or stimulating personal growth, learning, and development¹. Examples are team support and job autonomy. Job demands and job resources create two processes: a deteriorating health process and a motivational one. Besides, job resources can cushion the impact of job demands on stress, and they mainly influence motivation when job demands are high.

As the theory matured, other propositions advanced¹. For example¹, identified personal resources, which relate to individuals' beliefs about how much control they have over their work environment and occupational resources. Thus, optimistic individuals with high self-efficacy are found to believe that good things will happen to them and that they can handle the unexpected, triggering motivational processes.

One of the reasons for the widespread use of JD-R theory is that it predicts the crucial state of mind of the worker. For example, the theory explains the generation of burnout, work engagement, apathy, and boredom¹. Furthermore, the theory predicts job performance through the psychological mechanism of motivation and strain. Motivation positively impacts job performance, while job strain negatively affects job performance. Employees motivated by their work will likely engage in job-crafting behaviors, leading to higher levels of job and personal resources and motivation¹⁹. Simultaneously, employees stressed about their jobs are prone to self-undermining behaviors, leading to higher levels of job demands and even higher levels of job stress¹.

In a recent study²⁰, have expanded on the JD-R theory to understand how organizations and employees can navigate crises. They argue that looking at job demands and resources cannot explain individuals' well-being and motivation during challenging times. Instead, their research introduces 17 propositions considering the interaction between job demands and resources from different perspectives, such as the individual, family, job, and organization. These propositions also highlight the importance of various factors, including individual

regulatory strategies, family support, leadership, teamwork, and organizational support in shaping job-related outcomes.

Episodic perspective

The Episodic Process Model (EPM) connects emotional experiences with individual performance within specific time-bound episodes²¹. These episodes are characterized by coherent thematic organization and are influenced by goal structures. Performance episodes, a key concept in this model, are thematically organized segments of work activities related to immediate goals or desired end-states. They are essential to the model as they structure individual performance and are subjectively experienced.

Episodes are distinct from tasks and are bound by time. Unlike tasks that can be resumed, they end when the desired end state is reached or when the goal is abandoned. While some behavior episodes may not be performance episodes, as they lack organizational relevance, they play a significant role in daily work experiences.

Attentional focus is crucial within performance episodes, with cognitive resources being essential for effective performance. People bring various resources, including skills and knowledge, to their tasks. The allocation of these resources affects task performance, with the ability to focus attention on the task being a critical factor²². Affective states influence episodic performance by creating attentional demands that divert focus away from the task. Distractions, interruptions, and off-task thoughts can negatively impact performance by consuming cognitive resources. Successful regulation, or the ability to stay on task, depends on individual regulatory resources and the task's attentional pull^{23,24}. Regulatory resources may vary between individuals and throughout the day, affecting their ability to stay focused on tasks. Factors like task importance, intrinsic motivation, and goal specificity influence attentional focus. Challenging goals and intrinsic motivation enhance performance by directing attention toward the task²⁵.

Moreover, episodic fluctuations in job demands and resources can impact task absorption. Job demands can be categorized as challenging or hindering, with hindrance job demands negatively affecting task absorption. Job resources play a role in mitigating the negative effects of job demands on task absorption, especially in the presence of challenging demands. The Episodic Process Model links emotional experiences with performance within specific episodes of time-bound, goal-driven work activities. Attentional focus and allocating cognitive resources are essential for effective performance within these episodes, which job demands and resources can influence.

Hypotheses development

Job resources refer to those physical, psychological, social, or organizational aspects of the job that are functional in achieving work goals, reducing job demands and the associated physiological and psychological costs, or stimulating personal growth, learning, and development^{1, 26}. It has suggested that job resources are associated with high levels of task absorption due to the cognitive effort and the increasing time the individual uses to make a decision¹. It suggests that a dynamic approach to job resources echoes other episodic processes, where the interpretations of specific events as they happen cause essential positive changes in task performance. Thus,

Hypothesis 1 *Episodic job resources are positively related to task performance.*

Task absorption is related to visual attention, concentration, and cognitive effort^{1, 27}. It argues that task absorption captures how individuals experience their work as something they want to devote time to. It is related to being immersed in their work activities that time seems to fly. Several studies using the JD–R framework have shown that task absorption at the general, day, and week levels is associated with task performance at the respective levels²⁸. Following the EPM proposition that performance is better, the more attentional and energetic resources are dedicated to the activity at hand, the more we can expect that there will be a positive relation between task performance and task absorption at the episodic level. Hence, examining how much task absorption relates to task performance throughout different work activities is crucial. A recent eye-tracking lab study showed that intermediate pupil diameter, which reflects task absorption, is positively associated with task performance in a visual letter n-back task¹⁰. Additionally, a research study on task absorption showed that students who were actively absorbed while solving matching problems had better task performance than students who were passively absorbed²⁹.

In sum, few studies have shown that task absorption and task performance are related at the episodic level; nevertheless, empirical evidence for the relation between task absorption and task performance at the episodic level is still lacking. Thus, we will examine this relation in a decision-making episode to test the episodic task absorption–task performance link. We propose that absorbed individuals will achieve higher task performance during a task by using all their cognitive effort. Additionally³⁰, it states that critical for effective functioning at work is when employees keep their cognitive effort to invest the right resources during an episode, and¹ mentions that job resources can trigger task absorption. On this basis, we propose,

Hypothesis 2 *Episodic job resources are positively related to task absorption.*

Job demands are those physical, psychological, social, or organizational aspects of the job that require sustained physical and psychological effort and are, therefore, associated with specific physiological and psychological costs²⁰. Examples are high work pressure and emotionally demanding interactions with clients or customers. Following¹³, who reported that task absorption is related to biomarkers like pupil diameter, we aim to provide a more precise and comprehensive understanding of task absorption during an episode and deepen knowledge about their relationship with job demands/resources through pupil diameter analysis during a decision-making

episode. The decision-making episode occurs on a small scale and repeatedly during the task's duration. Assuming this perspective, an individual might feel engaged at one point and exhausted at another. In addition, high job demand and high job resources can generate motivation¹, while other constellations of job demands/resources can be related to strain during decision-making. Then,

Hypothesis 3 *The relationship between episodic job resources and task performance is mediated by task absorption.*

The EPM outlines how cognitive processes during work activities are theoretically related to task performance. Specifically²¹, postulates that for individuals to perform well, their cognitive effort needs to be entirely focused on the task. As mentioned in this research, task absorption refers to cognitive effort during a task. Task absorption draws a person's cognitive effort toward it, facilitating the self-regulation of attention toward it⁵. We propose that task absorption is a central factor in understanding how episodic job demands/resources influence task performance in decision-making. Specifically, job demands/resources are described as a distal antecedent of task performance, such that this first relates to task absorption, which leads to better task performance. The mediating function of cognitive effort and individual behavior has been widely acknowledged in theory and research¹, emphasizing that cognitive effort offers individuals relevant information through improving cognitive effort during an episode²¹. Furthermore, the theory of task absorption has highlighted the absorbed effect (e.g., making individuals feel with strong attention) of support for optimal task performance²⁷. Therefore,

Hypothesis 4 *The indirect effect of episodic job resources on task performance through task absorption will be moderated by episodic job demands.*

To expand on previous research, we adopted the JD-R theory¹ to describe and test a mediation model where task absorption, measured using individuals' pupil diameter, represents a critical element of fostering task performance in a decision-making episode. Figure 1 shows our hypothesized model.

Materials and methods

Design

This research employs a within-subjects design, where all participants are exposed to each condition of the independent variable. We chose this design to reduce variability. Since the same participants are used in all the four conditions, individual differences are minimized, leading to greater statistical power. Additionally, this design often requires fewer participants because each one experiences multiple conditions. Finally, the increased sensitivity in this design allows differences in task performance to be more easily attributed to the experimental manipulation of job demands and job rather than individual differences³¹.

Convenience sampling was utilized, with participants probabilistically assigned from a list of potential candidates. The invitation was sent via press releases and email invitations, using contact information obtained from a Master of Business Administration (MBA) alumni database. Exclusion criteria included individuals with visual impairments and managers with a minimum of five years of experience.

Apparatus

The apparatus consisted of a questionnaire, an eye-tracking device, and a computer showing information and registered the decisions. The stimuli were presented on the screen of a computer using a program writing in Python v.3.7.0. Ocular behavior and pupil diameter were recorded binocularly with an eye-tracking device (Eye Link 1000[®]) at a 500 Hz sampling rate. The EyeLink eye-tracker can detect changes in pupil diameter as small as 0.1%, and its high sampling rates allow for precise measurement of pupil responses. A neuroscience assistant supervised the 10-minute tutorial, connected and calibrated the eye-tracker, and verified all experimental conditions. Subjective data were collected weekly through an electronic survey when participants attended the laboratory experiment. Initially, they completed a demographics questionnaire, followed by the psychological capital scale. The assistant then provided an iPad for the survey, conducted via the Qualtrics XM software version 3.2.0 (<https://www.qualtrics.com/>), which automatically collected participants' responses. The survey included

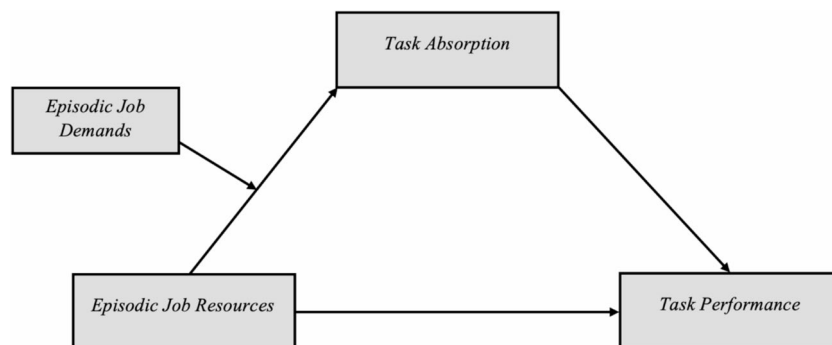


Fig. 1. Hypothesized model.

informed consent, demographic questions, and four scales with numerical (1 to 5) and verbal (“Strongly disagree” to “Strongly agree”) anchors. Participants then executed the experimental task.

Participants

The University’s Ethics Committee at the University of Chile approved the experiments under registration number 003/2021. All experiments were conducted in accordance with relevant guidelines and regulations. Participants provided written informed consent, which was also approved by the University’s Ethics Committee. The committee’s guidelines and regulations comply with the Declaration of Helsinki.

We contacted managers with at least five years of work experience. The recruitment period for this study began on April 1, 2021, and concluded on January 30, 2022. We initially selected 19 participants but collected valid data from 18 participants, as one was discarded due to issues with the eye-tracker. The final sample comprised 18 healthy managers aged 28 to 65 years ($M = 35$ years, $SD = 5.87$) from various companies in Chile. The participants included 14 males and four females, with an average work tenure of 10.2 years ($SD = 5.45$) (Table 1). Their occupations were business/management professionals (43.7%), civil engineers (28.1%), sociologists, psychologists, political scientists (15.3%), and others (lawyers and military; 12.9%). Participants were from organizations within the services (50.4%), financial and accounting (30.6%), information technology (18.1%), and other (5.9%) economic sectors.

At the time of the study, all participants were executive managers, carefully selected for their consistent background in business knowledge and organizational experience, as evidenced by their MBA degrees. We intentionally excluded top management teams, corporate executives, company founders, low-level managers, and first-line supervisors to focus on individuals with relevant skills and experience in resource administration. Additionally, no participant was enrolled in any course during the experiment.

Participants were compensated based on their performance percentile in the task, with benchmarks set by a prior pilot study involving 40 engineering students. Compensation was tiered as follows: \$10 for performance below the 50th percentile, \$15 for the 50th to 75th percentile, and \$25 for performance above the 75th percentile. Participants were informed in advance that their performance could lead to different rewards, which might have affected their motivation. To address this, we factored in the managers’ experience and salary levels. Since the managers were already highly compensated, the performance bonus represented only a small percentage of their monthly income, minimizing its influence on their motivation and behavior. Additionally, we conducted the experiment on different days for each participant, further reducing potential biases. The final sample included 18 participants, with no exclusions or missing data.

Procedures

We ensured all participants were well-rested and in good health, as confirmed by self-report. Participants reported sleeping for seven or more hours and were asked to abstain from caffeine and alcohol for 24 hours before the experiment. They provided written informed consent. All participants had normal vision, reported no psychiatric or neurological conditions, and did not consume drugs affecting the nervous system. These requirements were crucial as they could influence pupil diameter. The risks associated with the survey data collection and eye-tracking procedures did not exceed possible discomforts or dangers like those experienced daily³². Eye-tracking analysis designs have no contraindications or risks as it is a non-invasive procedure. Given this, the measurement procedure employing eye-tracking analysis does not threaten physical and psychological health or cause discomfort. A trained researcher was present to interrupt the procedure if abnormal behavior was detected in the participants.

Given the face-to-face data collection and the ongoing pandemic context, we implemented safety protocols to protect participants against potential contagion. The day before the experiment, participants were required to submit a health passport form, detailing their symptoms and providing contact information for household members to facilitate traceability. The lead researcher minimized direct contact with participants and underwent a PCR test to ensure they were not carrying COVID-19. Participants were instructed to maintain a physical distance of 2 m from the researcher throughout the experiment. This study received ethical approval from the Human Research Ethics Committees of both the Faculty of Medicine and the Faculty of Economics and Business at the University of Chile.

A 3 × 3 m laboratory with blackout curtains housed the experiment. Participants were seated in a darkened space with their heads resting comfortably on a head stabilizer. A camera recorded the participants from in front of the screen. A tutorial programmed on the computer trained participants before the simulations. Stimuli were presented on an LED monitor (ViewSonic, 27”) 52 cm from the subject, covering a visual field of 33° x

Variables	M	SD	1	2	3	4	5
1. Gender	0.74	0.45					
2. Age	35.1	5.87	0.21*				
3. Work experience	10.2	5.45	0.38*	0.90*			
4. Psychological capital	4.34	0.47	0.11	0.10*	0.02*	(0.90)	
5. Pupil diameter	0.04	1.10	0.45	0.73*	0.12**	0.24*	
6. Task performance	0.12	1.01	0.39*	0.32*	0.29*	.26**	0.73**

Table 1. Means, Standard Deviations, and correlations. Reliabilities are in parentheses. * $p < .05$; ** $p < .01$.

49.5°. Calibration adjustments were performed before starting the experiment, during which participants were instructed to explore freely.

We designed four conditions to study different job demand/resource combinations: (1) Condition 1: Low job demand/Low job resource; (2) Condition 2: Low job demand/High job resource; (3) Condition 3: High job demand/Low job resource; and (4) Condition 4: High job demand/High job resource. Figure 2 shows the established conditions for this experiment (Fig. 2).

The task

We used the Beer Distribution Game to set up the experiment with the software Presentation[®]. The Beer Distribution Game is a role-playing simulation of an industrial production and distribution system developed at the Massachusetts Institute of Technology to introduce management students to business dynamics and computer simulation. It has been played by thousands worldwide over nearly six decades, ranging from high school students to chief executive officers (CEOs) and government officials³³. We programmed a computational interface that allowed a single participant to play the beer game with eye-tracking analysis, where the incoming order matched precisely with the simulated demand. As mentioned, this experiment measured response times in milliseconds. When the fixed stimuli appeared, participants had a free vision of the screen, which was split into nine regions of interest (ROIs) during each decision round.

The Initial Inventory was the number of units in the inventory during the first trial. The Incoming Inventory refers to the incoming delivery order. The Final Inventory was calculated as the Initial Inventory plus the Incoming Inventory minus the order. The demand was the incoming order, and the number of shipments was computed automatically. The order decision value remained empty until the participant typed a number, after which they pressed the “enter” button to proceed to the next screen with information for the subsequent decision. After each ordering decision, the algorithm displayed a fixation cross at the center of the screen. Figure 3 shows the screen layout for each trial.

The sequence of screens and stimuli was randomized across participants. The information displayed was dynamically adjusted based on each participant’s decisions; for example, if a participant requested more stock units, the on-screen data would update accordingly.

Subjective, behavioral, and physiological measures

We gathered data using subjective, behavioral, and physiological measures. Each participant’s levels of job demand and job resource were assigned based on their task decisions. Although the demand function was uniform across all subjects, we introduced slight variability at each step. We collected eye-tracking data from all participants and analyzed it across four conditions. To streamline this process, we created a dataset titled “conditions by subject,” with conditions as columns and subjects as rows.

Subjective Measures. We used a self-reported psychological capital scale to add additional control variables. Psychological capital consists of personal resources such as hope, efficacy, optimism, and resilience. Previous research has highlighted its importance in general workplace performance³⁴. According to³⁵, psychological capital is an individual’s positive psychological state of development, characterized by: (1) self-efficacy, or the confidence to take on and put in the necessary effort to succeed at challenging tasks; (2) optimism, or making a positive attribution about succeeding now and in the future; (3) hope, or persevering toward goals and redirecting paths to achieve them when necessary; and (4) resilience, or sustaining and bouncing back from adversity to attain success³⁵. We used the Spanish version of the scales for hope, self-efficacy, resilience, and optimism³⁴. In the next section, we report the statistical analysis following the criteria for conducting organizational research suggested by³⁶.

Behavioral Measure. The behavioral measure in this experiment was episodic task performance. Each participant received a score called “game performance,” used for statistical analysis. The inventory level determined the costs: a positive inventory value incurred a cost of 0.5 per unit, while a negative inventory value

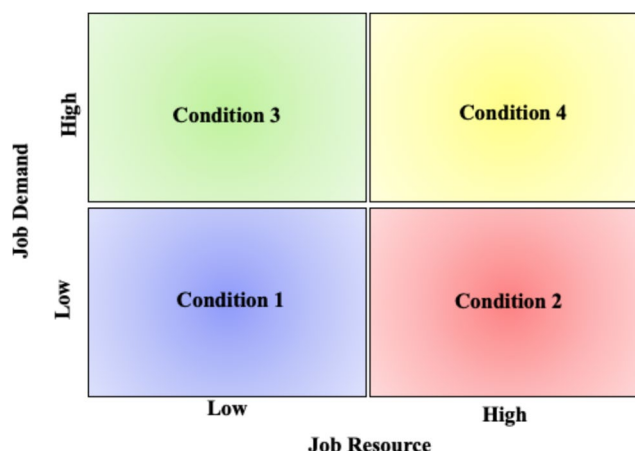


Fig. 2. Experimental established conditions.

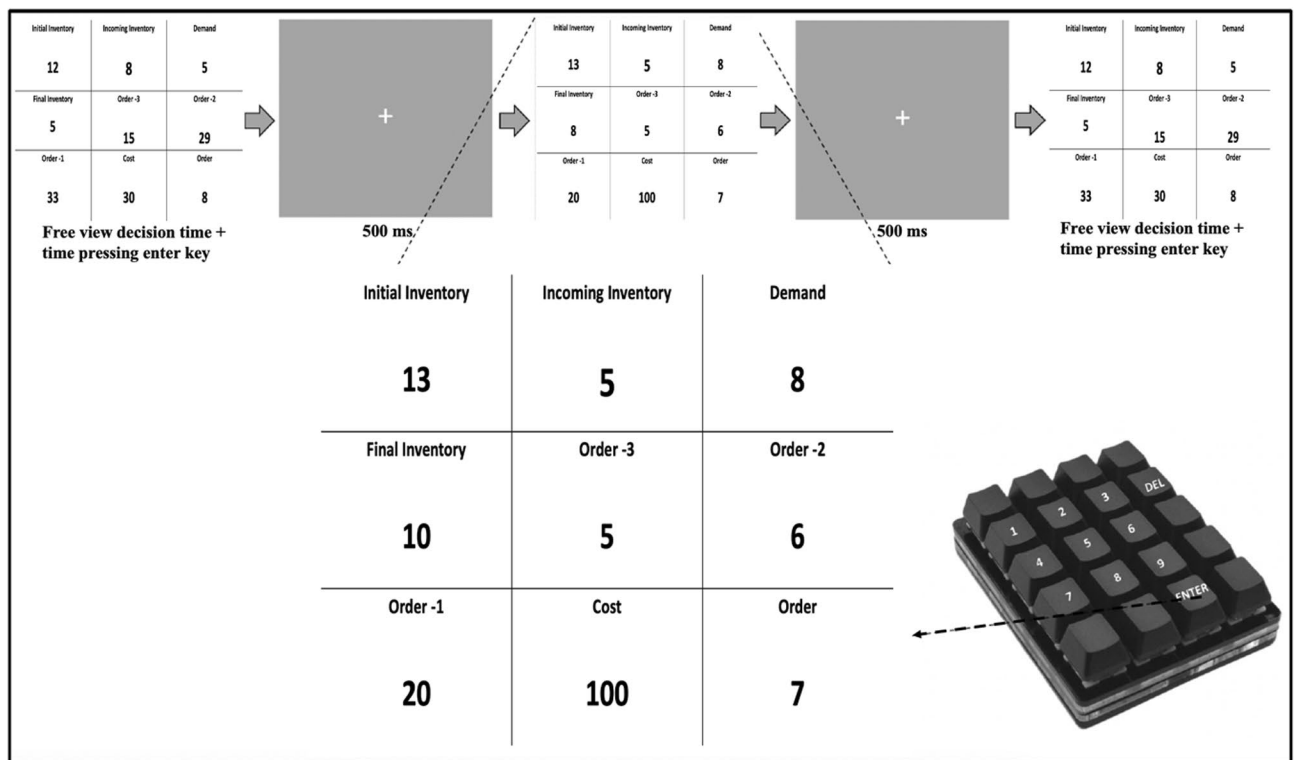


Fig. 3. The sequence of screens and stimuli during the task. The nine ROIs displayed on the screen for each trial were as follows: (ROI 1) Initial Inventory, (ROI 2) Incoming Inventory, (ROI 3) Demand, (ROI 4) Final Inventory, (ROI 5) Order – 3, (ROI 6) Order – 2, (ROI 7) Order – 1, (ROI 8) Cumulative Cost, and (ROI 9) The Amount of the Order.

incurred a cost of one per unit. Lower costs indicated better performance in the beer game, whereas higher costs indicated poorer performance. This score was inverted to facilitate analysis and interpretation of the results.

Physiological measure. The recorded ocular behavior and pupil data were processed and analyzed using custom-made MATLAB v.R2021a scripts. The pupil signal was processed by cleaning blinks with linear interpolation and applying a low-pass filter. Raw data were normalized to the z-score to control for pupil diameter variability among subjects³². Each trial was segmented from the beginning of the screen presentation to the end of the trial (i.e., when participants pressed the “enter” button) and included part of the fixation cross. The baseline was calculated as the average signal 500 ms before the transition. Participants viewed a fixation cross before all trials, which helps focus or divert attention to a particular screen position³⁷.

Controls. The survey included participants’ age, gender, and job experience as control variables since these variables can affect performance³⁸.

Results

Descriptive statistics

Means, standard deviations, correlations, and reliabilities of the variables are summarized in Table 1. Gender is a dichotomic variable (0 = male; 1 = female). Age was measured in years. Work experience was measured in years. As we mentioned, psychological capital is a score from a Likert-type scale. Task absorption is measured using pupil diameter in z-score, and task performance is a score. The task performance score is inverted, i.e., positive values indicate worse task performance, and negative numbers indicate better task performance. Internal consistency reliability was measured using Cronbach’s α . These alpha scores are acceptable for the analysis to follow the criteria of Podsakoff and Podsakoff (2019). These results provided support for internal consistency reliability.

Hypotheses testing

We modeled a first stage moderated partial mediation effect using Hayes’ Process macro to test our hypotheses³⁹. We used SPSS v.23 and Process macro v.2 16.23. Hypothesis 1 predicted a linear and positive relationship between episodic job resources and task performance (TP). In contrast, Hypothesis 2 predicted a linear and positive association between episodic job resources and task absorption (TA). Hypothesis 1 was supported for managers ($b = 0.67$, $SE = 0.04$, $p < .01$, model 2). Similarly, Hypothesis 2 was also supported ($b = 0.35$, $SE = 0.01$, $p < .01$, model 1). Hypothesis 3 predicted a mediation effect of task absorption on episodic job resources and task performance. To test the mediation effect of Hypothesis 3, we made the independent and mediation variables the predictors of the dependent variable (Model 3), calculated regression coefficients, and estimated the mediation

			Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
			Dependent Variables					
Independent Variables	R ²	R ² Adj.	TA	TP	TP	TA	TA	TP
Episodic Job Resources	0.12	0.10	0.35**	0.67**	0.33**	0.19**	0.10**	0.36**
			(0.01)	(0.04)	(0.04)	(0.01)	(0.01)	(0.04)
Task Absorption	0.19	0.18			1.29**			1.19**
					(0.09)			(0.09)
Episodic Job Demands	0.21	0.20				0.01**	0.01**	0.02**
						(0.01)	(0.02)	(0.06)
Episodic Job Resources × Episodic Job Demands	0.25	0.24					0.06*	0.07
							(0.01)	(0.05)

Table 2. Regression results of Testing Moderated and Mediation effects. The table shows the regression results for the six tested models. Standard errors are in parentheses. * $p < .05$; ** $p < .01$.

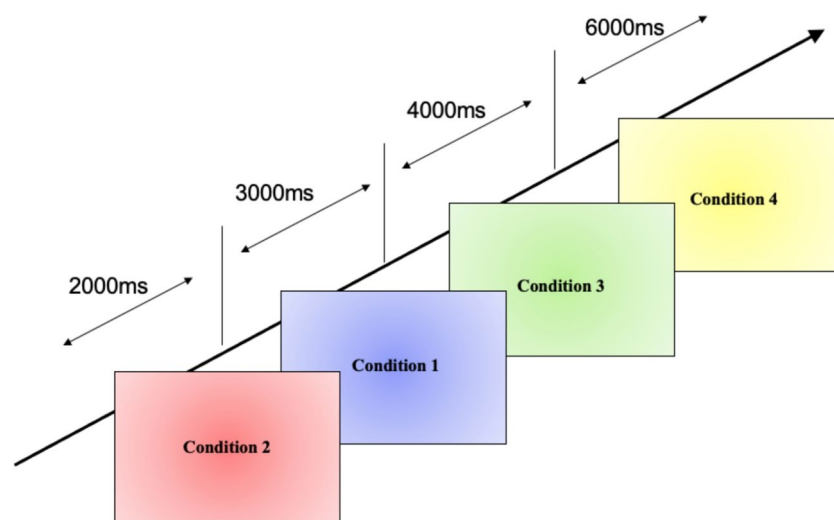


Fig. 4. Representation of the time of the decision.

effect. The mediation effect test was significant (Estimate = 0.30, SE = 0.03, bootstrapped 95% CI: [0.24, 0.37]). Hypothesis 3 was supported. Hypothesis 4 predicted a moderation effect of episodic job demands on the relationship between episodic job resources and task absorption. As Table 2 shows (Model 5), the interaction effects were significantly different from zero for managers ($b = 0.10$, SE = 0.01, $p < .05$, Model 5).

Further analysis

To ensure that task absorption differs from each condition, we performed an ANOVA on the decision time [$F(1,108) = 4.71$, $p = .032^*$]. * $p < .05$. Figure 4 includes the time of the decision for the four conditions. As we expected, decision times are consecutively ordered. In condition 2, the decision time is shorter than in condition 1. In condition 1, the decision time is shorter than in condition 3. In condition 3, the decision time is more straightforward than in condition 4. The total task average trial time was 5000 ms. Nevertheless, the trial's average duration that follows the condition was 6000 ms. In condition 1, the decision was made after 1800 ms. In condition 2, the decision was made after 2900 ms. Condition 3 lasted 3900 ms. The most prolonged decision trial duration was condition 4 with 6000 ms.

Figures 5 and 6 using MATLAB. The pupil diameter dynamics plot has the standard deviation between subjects shaded around the mean curve. Even though the results are insignificant, there is a clear tendency for a variation between the four conditions regarding pupil diameter. To ensure that attention differs from each condition, we performed an ANOVA on the pupil diameter [$F(1,104) = 7.52$, $p = .541$]. Exploring the four conditions in the first 1000 ms (Fig. 5), we found that immediately after the transition of each condition, the pupil exhibits a fast dilation that reaches its most considerable value around 500 ms. This response was followed by a contraction phase, with the smallest value near 1000 ms for all the conditions. Figure 5 shows the normalized average pupillary response curve according to the conditions the subject transits.

The average normalized curve of pupillary response according to the four conditions is shown in Fig. 6. I found that the pupillary answer studied was influenced by the level of job demands and job resources, where differences exist from the contraction to the dilation phase. Pupil diameter is shorter when the individual is in a low job demand/low job resources condition (Condition 1). Pupil diameter is shorter when the individual is in

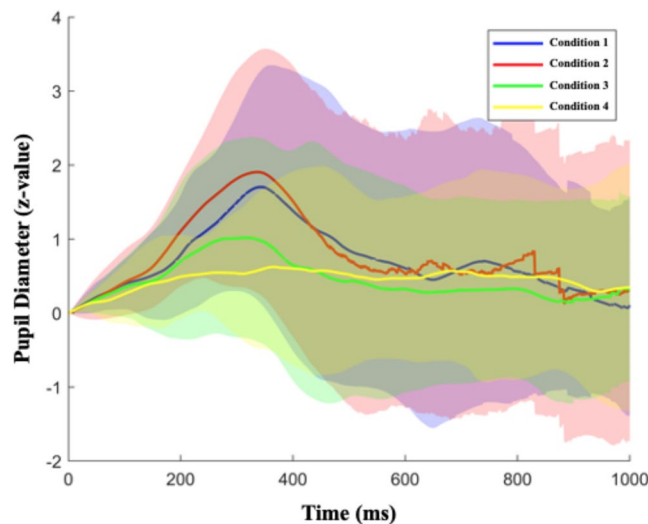


Fig. 5. Pupil diameter by conditions in the first 1000ms.

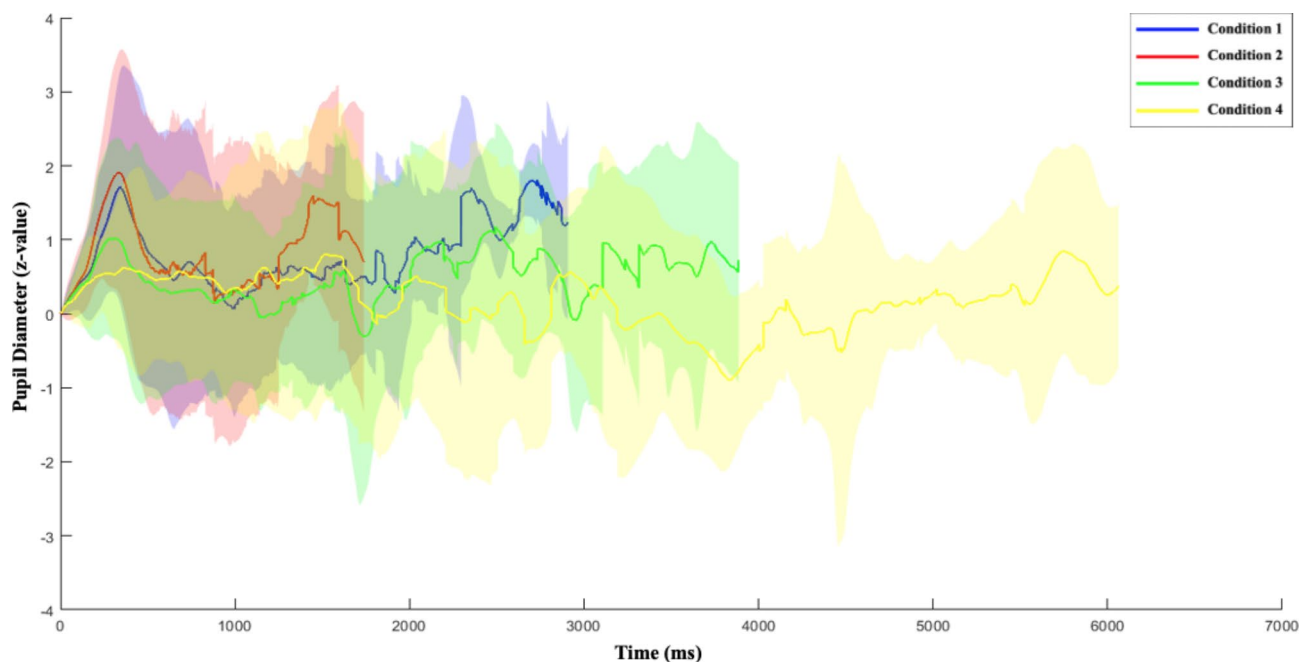


Fig. 6. Pupil diameter and time decision by conditions.

a low job demand/high job resources condition (Condition 2). Pupil diameter is larger when the individual is in a high job demand/low job resources condition (Condition 3). Pupil diameter size is more prominent when the individual is in a high job demand/high job resources condition (Condition 4). The pupil diameter varies across the four conditions. Interestingly, condition 4 is where the participants took the longest to make the decision.

Discussion

Our work integrates the JD-R theory¹ and the EPM⁵. We present a study aimed to test whether cognitive mechanisms, such as task absorption, are related to task performance through the experience of episodic job demands and resources. We examined these associations at the episodic level while accounting for the effects of job demands and job resources on performance through task absorption. The results suggest that the relationship between episodic job resources and task performance is mediated by task absorption. Individuals perform better when they feel more absorbed in a work activity such as a decision-making task. The results support our four hypotheses, namely episodic job resources are positively related to task performance (H1), episodic job resources are positively associated with task absorption (H2), the relationship between episodic job resources

and task performance is mediated by task absorption (H3), and episodic job demands moderate the relationship between episodic job resources and task absorption (H4).

Our findings have critical theoretical and practical implications. First, we state that task absorption is a central factor in understanding how episodic job resources influence task performance in a decision-making task. We demonstrate that task absorption will mediate the relationship between episodic job resources and task performance. Furthermore, for the mechanism of task absorption, the results of our study indicate that when managers experience high levels of task absorption, they are more immersed and have high levels of cognitive effort during the decision-making task. This aligns with the EMP²¹ absorption, which deploys better attention and increases performance.

Second, we employed the Beer Distribution Game and explored the dynamics of visual attention using an eye-tracking technique to collect pupil diameter measurements. We empirically analyzed pupil diameter during a repetitive decision-making episode¹⁰. We investigated whether changes in demand and resources were related to changes in baseline pupil diameter. The novel method can give insights to better sense and preemptively theorize the future. Our research delivers empirical evidence that affective outcomes of job demands, and resources might be related to profiles of functioning in biological systems and may be relevant for the risk of development of physical illness.

Third, the results of this study can be used for implementing decision-making-focused skills development training in which individuals become aware of their decision-making pacing styles and the importance of breaks when performing a decision-making task^{40, 41}. Furthermore, this research highlights the significance of physiological measures that can complement self-reported measures and provide a more objective indicator of task absorption when companies look for data about their managers. A multidisciplinary team can create training with neurofeedback. For example, neuro-strategy training programs teach managers or employees to avoid problems that can trigger task absorption. Additionally, the insights of the present study can be used for individual time management training programs in which individuals become aware of their pacing styles and how to effectively manage the consequences of a particular type for one's level of task absorption and, consequently, one's task performance. Additionally, companies must promote a wide range of personal resources, like psychological capital, to develop empowerment managers with better task performance.

Although the present study adds further insight into task absorption during an episode, some limitations should be considered. Specifically, a large sample size will be significant for another study. Furthermore, designing a different task to test the hypotheses could benefit future research (e.g., bottom-up-based tasks, other job demands, and resources). Another thing that could be seen as a limitation is that we chose to adopt a within-subject design to examine our research aims. An essential strength of the within-subjects is minimizing error variance and the influence of individual differences in reward evaluation. Although I have no specific reason to expect that a study replication with a between-subjects design would yield very different results, we encourage scholars to pursue such a replication that would extend our present contribution.

We opted not to use repeated measures or multilevel mixed methods analysis in this instance due to the specific design and objectives of our study. Our focus was on comparing overall trends and differences between conditions rather than examining individual variations across conditions. However, we recognize the potential value of such analyses for future research. Other limitations are related to binocular rivalry. We elaborate on stimuli dominance but simultaneously focus on visual stimuli of the task at hand. Factors that divert attention from the task are often not inherent to the task but are part of the surrounding environment. Future studies must explore the role of neurobiological factors such as binocular rivalry or interference.

One strength of this work was using a sample of individuals and the best measurement instruments and processes available in a neuroscience laboratory. Furthermore, considering an episode approach, we use an incentive to improve ecological validity. However, future studies can improve ecological validity with portable instruments. In this line, executing experiments in the workplace in an authentic work environment can benefit the analysis and results of any future work. Future studies can better predict how managers perform in a digital era using cognitive neuroscience methods. Finally, further investigations based on other biomarkers of task absorption can be crucial to increase knowledge about physiological data and task absorption. For example, electroencephalogram (EEG) analysis explores whether task absorption is related to cerebral activity (e.g., theta and alpha waves, P300). Additionally, studies can add other subjective measures like control variables (e.g., skills and other personal resources).

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Author contributions

J. J. Design study, data acquisition, analysis and interpretation of data. Drafting and critical revision of the man-

uscript. Final approval of submitted manuscript. Analysis and interpretation of data. J.T. Design study, data acquisition, analysis, and interpretation of data. Drafting and critical revision of the manuscript. Final approval of submitted manuscript. Analysis and interpretation of data. P.M. Design study and data acquisition. Final approval of submitted manuscript. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

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